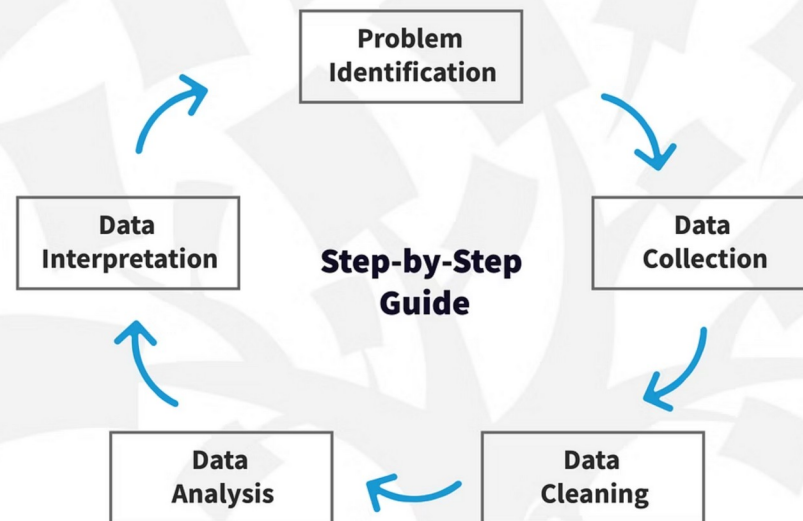


The Data Analysis Process



Data Analysis with Python

Case Study: Fruit Classification Dataset

A comprehensive exploration of data analysis through fruit classification, examining 20 fruit types with detailed physical and sensory features.

Group 9

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Dataset Choice and Problem Statement

The chosen dataset is the **fruit_classification**, carefully created to help learners and practitioners explore fruit classification using machine learning and data analysis projects.

It contains detailed information about **20 types of fruits**, each described with physical and sensory features such as size, shape, weight, color, and taste.

The main problem addressed is to analyze and understand how physical and categorical features can be used to differentiate and classify fruits, extracting insights and identifying patterns for potential classification or predictive tasks.





Data Understanding and Initial Inspection

01

Purpose

Understand the structure, content, and purpose of the fruit classification dataset before performing any analysis.

02

Codes Used

`df.head()`, `df.info()`, `df.shape`,
`df.columns`, `df.describe()`

03

Results

The dataset contains multiple numerical and categorical features describing fruits. The target variable is the `fruit_name` column. Initial inspection shows well-structured data with clear distinction between numerical and categorical variables.

Data Cleaning Process

Purpose

To ensure the dataset is consistent, accurate, and suitable for numerical and statistical analysis.

Codes Used

- `df.isna().sum()`
- `df.dropna(inplace=True)`
- `df.duplicated().sum()`
- `df[df.duplicated(keep=False)]`
- `df.drop_duplicates(inplace=True)`

Key Findings

The dataset contains **no missing values**, indicating excellent initial data quality.

All occurrences of duplicated rows were examined to understand the extent of repetition. The analysis removed **246 duplicate rows** from the dataset.

After cleaning, no significant duplicate observations were found, confirming the data is ready for analysis.

Exploratory Data Analysis (EDA)

To explore the dataset statistically and understand patterns and differences between fruit classes. EDA helps identify natural groupings and informs later classification logic.



1

Distribution Analysis

Analyzed fruit type distributions to highlight class proportions and potential imbalance, critical for classification tasks.



2

Shape Relationships

Explored relationships between shape and fruit type, revealing that certain shapes are strongly linked to specific fruits.



3

Numerical Variables

Examined weight and size to identify central tendencies, variability, and potential outliers in the dataset.



4

Economic Analysis

Analyzed average weight by shape and average price by fruit type, revealing meaningful relationships between physical characteristics and economic value.



Creation of New Analytical Metrics



Density Metric

$df['density'] = df['weight (g)'] / df['size (cm)']$

Identifies fruits that are heavy for their size, providing insight into fruit composition.



Price Per Gram

$df['price_per_gram'] = df['avg_price (\text{₹})'] / df['weight (g)']$

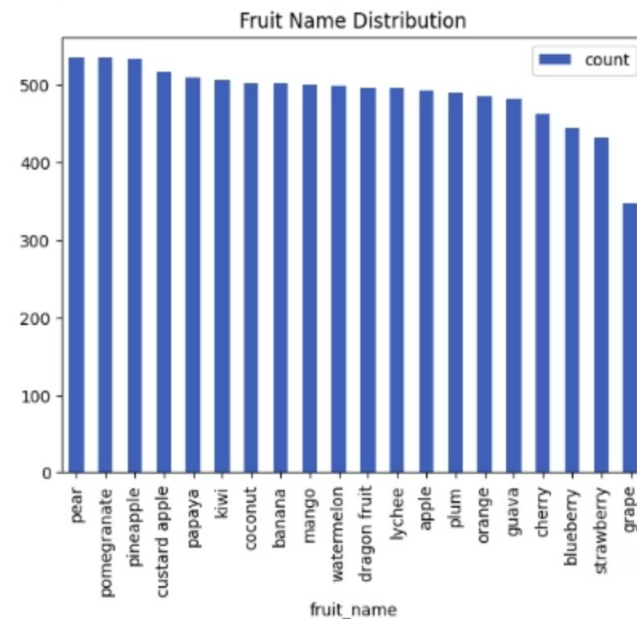
Reveals which fruits are more expensive per gram, enabling better comparative analysis.

These new metrics enrich the dataset by providing deeper insights and may enhance classification performance in machine learning models.

Data Visualization Insights

Bar Chart: Fruit Distribution

```
8]: Text(0.5, 1.0, 'Fruit Name Distribution')
```



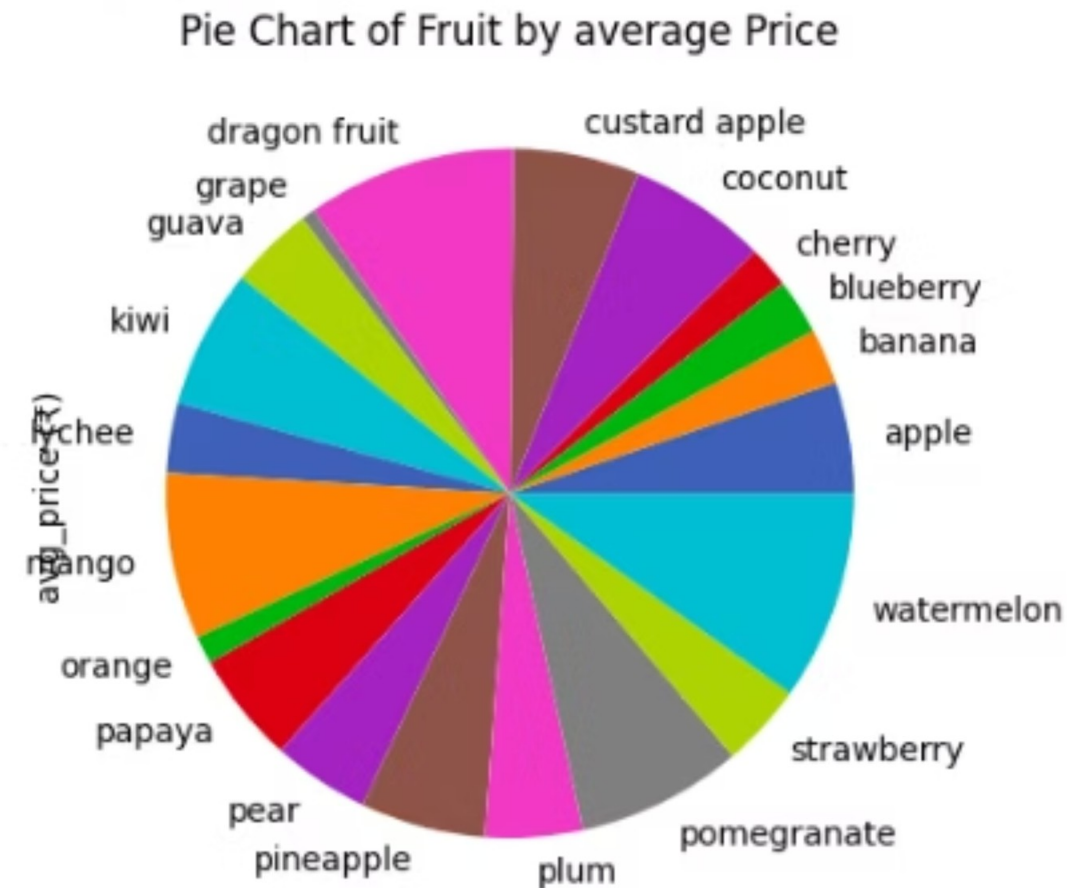
The bar chart shows the distribution of observations across different fruit types.

Well-represented fruits: Pear, pomegranate, apple, and pineapple have the highest number of samples.

Underrepresented fruits: Grape, strawberry, and blueberry appear less frequently.

This indicates the dataset is not perfectly balanced across fruit classes, requiring special attention to avoid bias in classification models.

```
[81]: Text(0.5, 1.0, 'Pie Chart of Fruit by average Price')
```



Pie Chart: Average Price

The pie chart represents the proportion of average price contributed by each fruit type.

Higher-priced fruits: Watermelon, pomegranate, and pineapple occupy larger portions, indicating higher average prices.

Average price varies significantly between fruit types, influenced by factors such as size, weight, and possibly market demand or rarity.

Project Conclusion

1	Initial Inspection Understood the structure and purpose of the dataset
2	Data Cleaning Ensured data consistency and reliability by removing 246 duplicates
3	EDA Identified important patterns in fruit distribution, physical characteristics, and prices
4	New Metrics Created density and price-per-gram metrics for deeper insights
5	Visualization Made patterns clearer and easier to interpret through charts

How do physical characteristics (size and weight) and fruit type influence the average price of fruits in the dataset?

The analysis shows that physical characteristics such as size and weight have a strong influence on fruit pricing. Larger and heavier fruits generally tend to have higher average prices, as observed in the scatter plots and group comparisons. However, fruit type also plays an important role, since fruits with similar physical characteristics may have different prices. Therefore, the average price of fruits is influenced by both physical attributes and fruit category.

Sources

- <https://www.kaggle.com/datasets/pranavkapratwar/fruit-classification>
- Python libraries: Pandas, NumPy, Matplotlib
- Course materials on Data Analysis

